Random variables and dispersion

Lecture and worked examples

Maths Explorer

Chapter path

Objectives. Build a distribution; interpret events; calculate expectation, variance, standard deviation; use linearity; simulate samples and calculate their observed mean.

Prerequisites. Probability, weighted mean, squares, trees and lists.

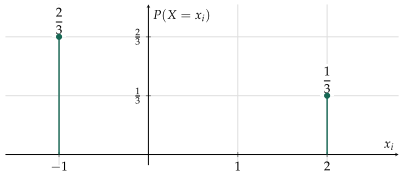
Yearly reference point. 19-28 January: 8 hours

Variable, event and distribution

On a finite sample space Ω, a real random variable X is a function that associates a real number with each outcome. The event $\{X = a\}$ groups the outcomes giving the value a. Its distribution associates with each distinct value x_i the probability $p_i = P(X = x_i)$, with $p_i \geq 0$ and $\sum p_i = 1$.

Variable, event and distribution

A fair die gives a gain X equal to 2 for 1 or 2, and to -1 otherwise. The distribution is $P(X = 2) = \frac{1}{3}$, $P(X = -1) = \frac{2}{3}$. $\{X \leq 0\} = \{3, 4, 5, 6\}$: probability $\frac{2}{3}$.



Variable, event and distribution

The values are not always equally likely even if the outcomes are. Group the outcomes that give the same gain.

Variable, event and distribution

Two fair coins are tossed. X counts the heads. Write its distribution and $P(X \leq 1)$.

Expectation and linearity

The expectation is $E(X) = \sum_i x_i p_i$. It describes the long-term theoretical mean, not a result that is necessarily possible on a single trial. For variables X and Y on the same finite sample space and real numbers a,b: $E(aX + b) = aE(X) + b$ and $E(X + Y) = E(X) + E(Y)$, with no assumption of independence.

Expectation and linearity

For the previous distribution $P(X = 2) = \frac{1}{3}$, $P(X = -1) = \frac{2}{3}$, $E(X) = 0$: the game is fair if X is the net gain. For $Y = 3X - 2$, $E(Y) = 3 \times 0 - 2 = -2$.

Expectation and linearity

An expectation is not a guarantee for an individual: the accumulated outcome can be a loss at every observed stage even when the expectation is positive.

Expectation and linearity

A gross gain is 10 with probability $\frac{1}{5}$ and 0 otherwise. What stake makes the game fair?

Variance, standard deviation and the König-Huygens formula

Writing $\mu = E(X)$, $V(X) = E((X-\mu)^2) = \sum_i (x_i - \mu)^2 p_i$. The standard deviation is $\sigma(X) = \sqrt{V(X)}$. The König-Huygens formula gives $V(X) = E(X^2) - E(X)^2$. A variance is always nonnegative; the standard deviation has the same unit as X, while the variance has the square of that unit.

Variance, standard deviation and the König-Huygens formula

For X with values 0 and 4 with probabilities $\frac{3}{4}$ and $\frac{1}{4}$, $E(X) = 1$, $E(X^2) = 4$, $V(X) = 3$ and $\sigma = \sqrt{3}$. Direct calculation: $(0-1)^2(\frac{3}{4}) + (4-1)^2(\frac{1}{4}) = 3$.

Variance, standard deviation and the König-Huygens formula

$E(X^2)$ is not $E(X)^2$. A variance calculated as negative signals an error or a very small rounding error to be dealt with explicitly.

Variance, standard deviation and the König-Huygens formula

For X with values $-1, 1$ equally likely, calculate the expectation, variance and standard deviation.

Transforming and comparing gains

For real a,b, $V(aX+b) = a^2V(X)$ and $\sigma(aX+b) = |a|\sigma(X)$. A translation moves the mean without changing the dispersion; a multiplication changes the scale. Two games with the same expectation can have very different dispersions.

Transforming and comparing gains

A game A always pays 2. A game B pays 0 or 4 with equal probability. Both have expectation 2; variances 0 and 4. Their distributions are therefore not interchangeable despite the same mean.

Transforming and comparing gains

The variance of a sum is not, in general, the sum of the variances; no rule of this type is used without hypotheses.

Transforming and comparing gains

$E(X) = 3$, $V(X) = 4$. Give the characteristics of $Y = -2X + 5$.

Samples and observed mean

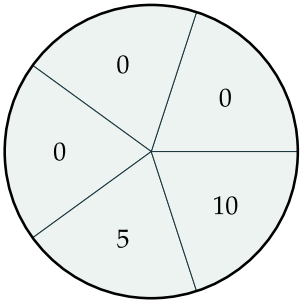
A sample of size n is here a list of n independent realisations of the same variable X. Its observed mean m varies from one sample to another; we compare it with $\mu = E(X)$.

Samples and observed mean

For X taking the values 0 and 2 with equal probability, $\mu = 1$. The sample 2, 0, 2, 2, 0, 0, 2, 2, 0, 2 has observed mean 1.2; another sample of the same size will usually give another mean. A simulation compares several observed means with μ without proving anything.

Samples and observed mean

Do not copy a single draw n times; do not confuse the observed mean of a sample with the expectation μ.

Samples and observed mean

X takes the values 0 and 2 with equal probability. Calculate the observed mean of the sample 2, 0, 2, 2, 0, then its gap from $E(X)$.